UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 004 · Implicit functions, problems on implicit and composite functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9 · S003·26	Deck 004 / 003, slide number (rendered .pptx)
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
T003b 00:10:33	Transcript 003 part II (implicit problems at its end)
CW U1 P17 · ESA Jan-24 s3 1d	Unit 1 class-work problem 17 · past-paper question

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5. Figure G1.1 · Level surfaces: spheres $x^2 + y^2 + z^2 = c$
6. Figure G5.1 · Polar coordinates and the inner derivatives

Colours follow the Okabe–Ito order: blue = main curve or first branch, vermilion = tangent or second branch, green and purple for further series. Dashed lines are tangents; grey dashed lines are references. The level-surface picture and the polar diagram are drawn from the equations discussed in the lecture; the slides' grapher screenshots are not copied.

1 Implicit functions, level surfaces and the differential

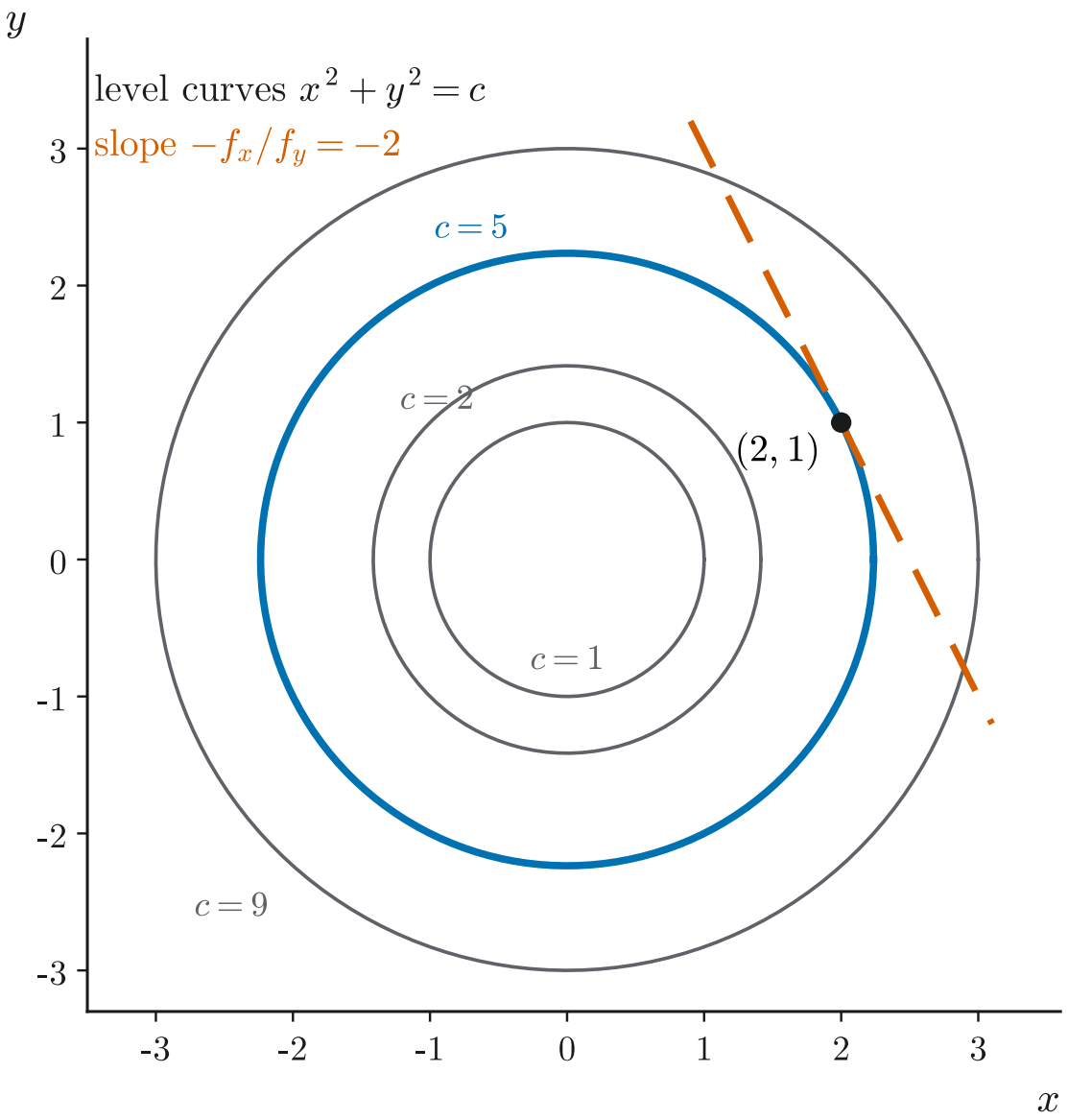


Figure G1.2. Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

How to read it. Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ you get by differentiating $x^2 + y^2 = 5$ directly.

$x^2 + y^2 + z^2 = c$: one shell per value of c

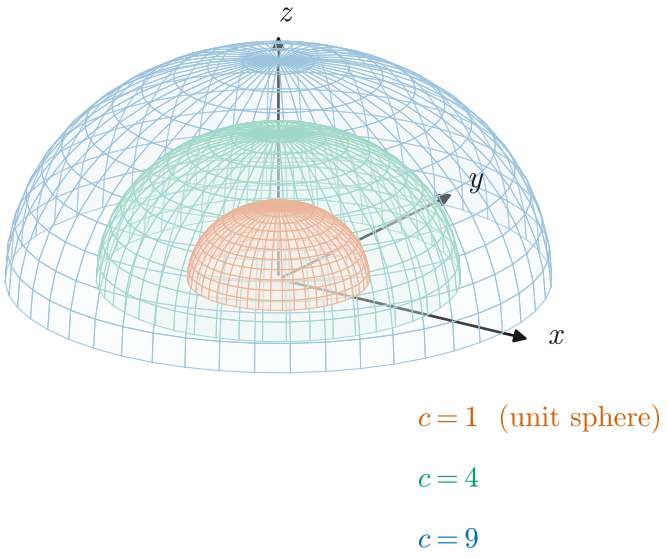


Figure G1.1. Level surfaces: spheres $x^2 + y^2 + z^2 = c$. ESA ★

S004·4 · T004a 00:03:29 · T004a 00:03:42

How to read it. Each shell is one level surface: on it $x^2 + y^2 + z^2$ has the same value c (only the upper halves are drawn, as on the slide). Larger c gives a larger sphere and $c = 0$ gives the origin. Think of each shell as one isotherm or equipotential surface. Moving along a shell does not change f, so $df = 0$ there.

2 Implicit differentiation in two variables

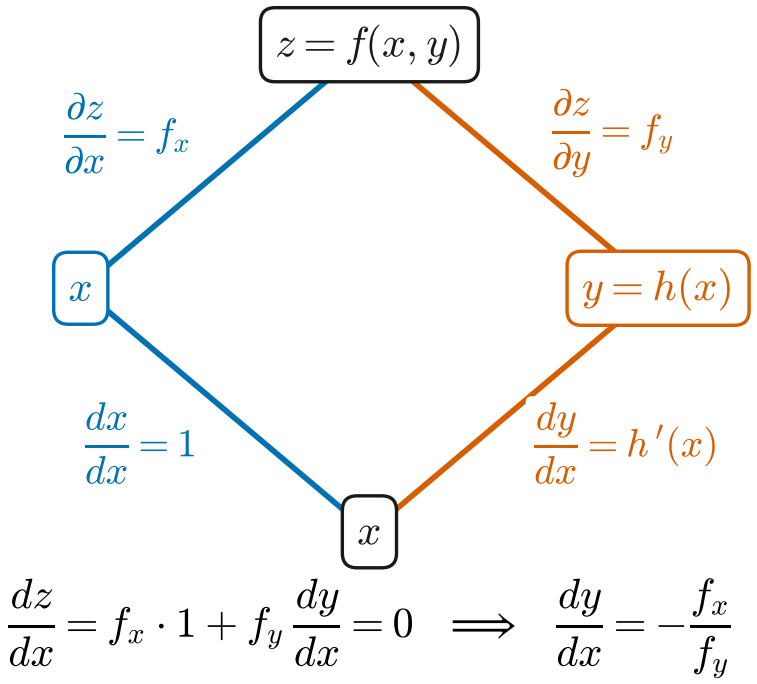


Figure G2.1. Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

How to read it. Differentiate $z = f(x, y)$ with respect to x along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y\, dy/dx = 0$.

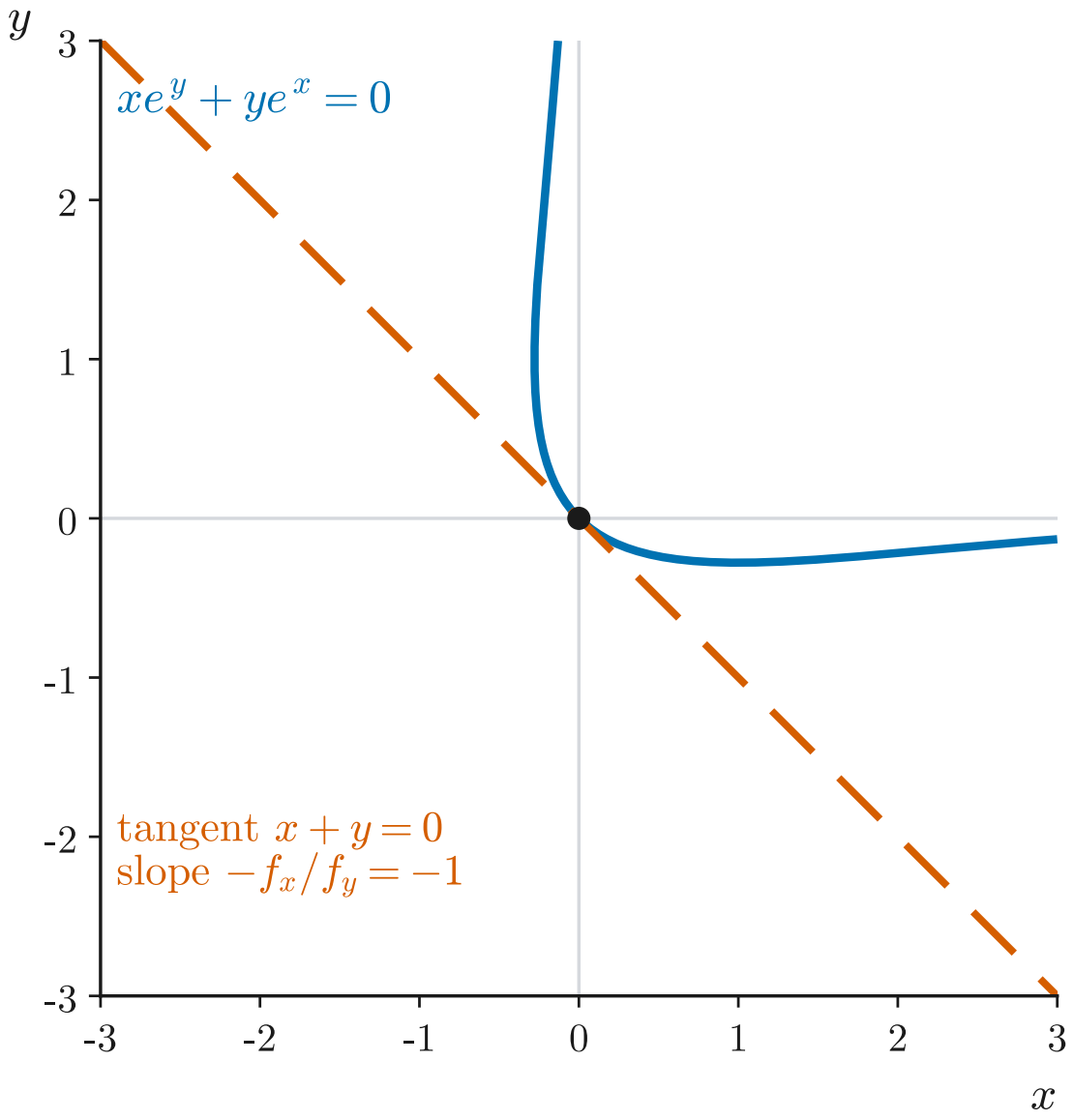


Figure G2.2. The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

How to read it. The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

4 Composite functions and the chain rule

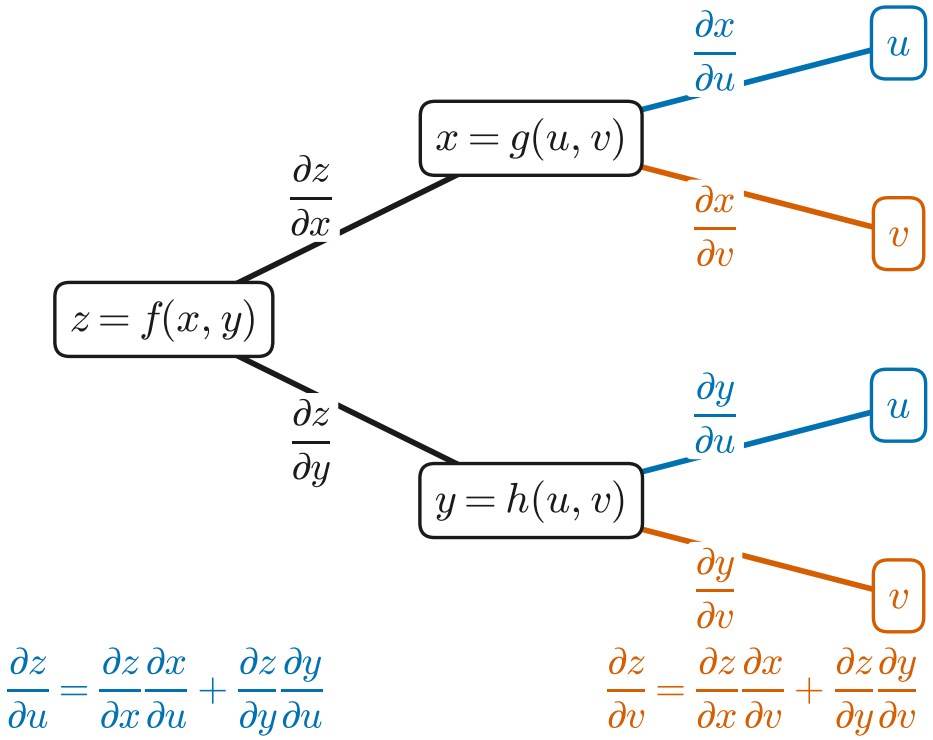


Figure G4.1. Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

How to read it. z branches to x and y, and each of those branches to both u and v, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at u (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at v (vermilion). In an exam, draw the tree first and read the terms off it.

5 Composite functions in polar coordinates

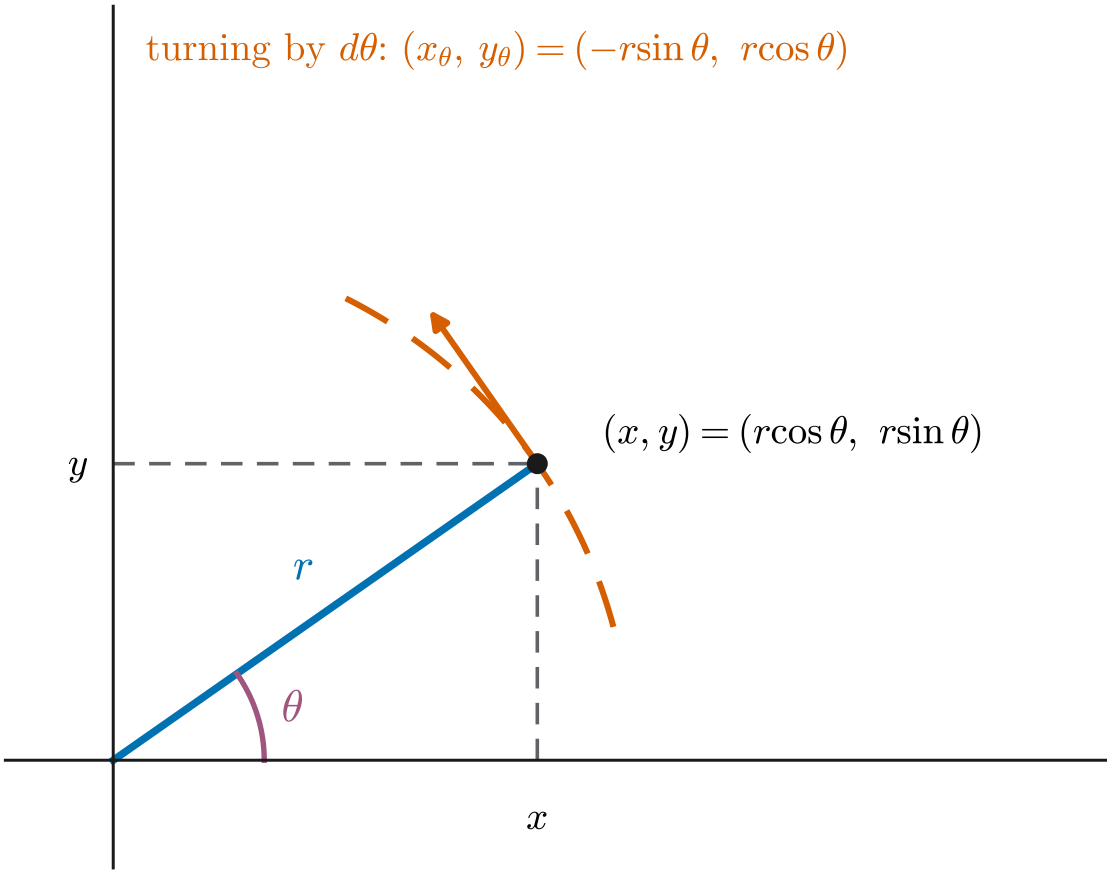


Figure G5.1. Polar coordinates and the inner derivatives. ESA ★★

ADDED FOR CLARITY S004·10 · S004·13 · S004·17

How to read it. A point at distance r and angle θ has $x = r\cos\theta$, $y = r\sin\theta$. Turning by $d\theta$ moves it along the dashed circle, so $x_\theta = -r\sin\theta$ and $y_\theta = r\cos\theta$, while r stays fixed. That is why any function of $x^2 + y^2 = r^2$ alone has $u_\theta = 0$.